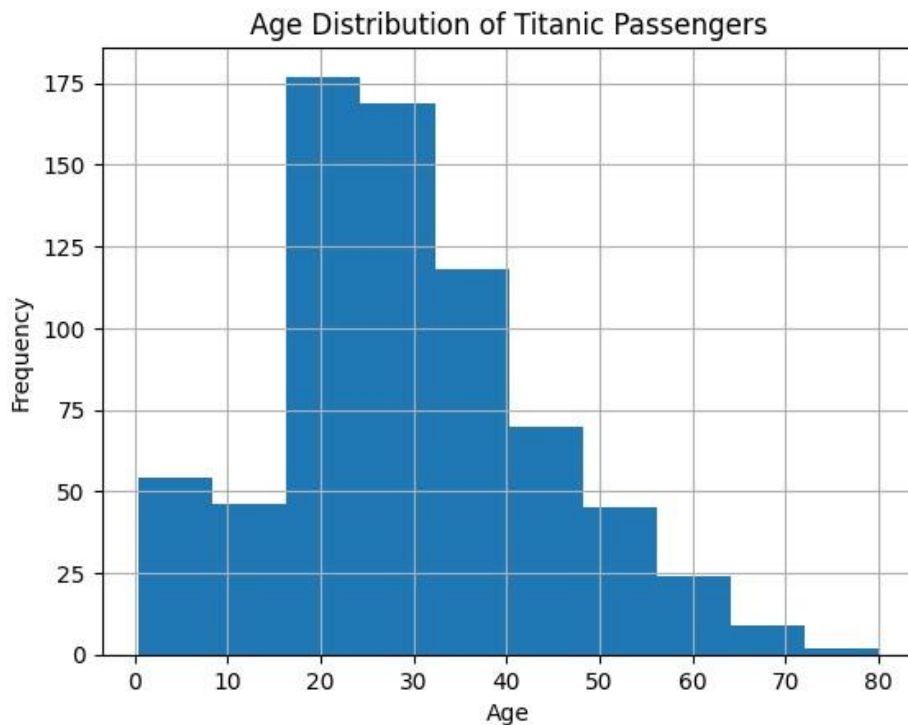


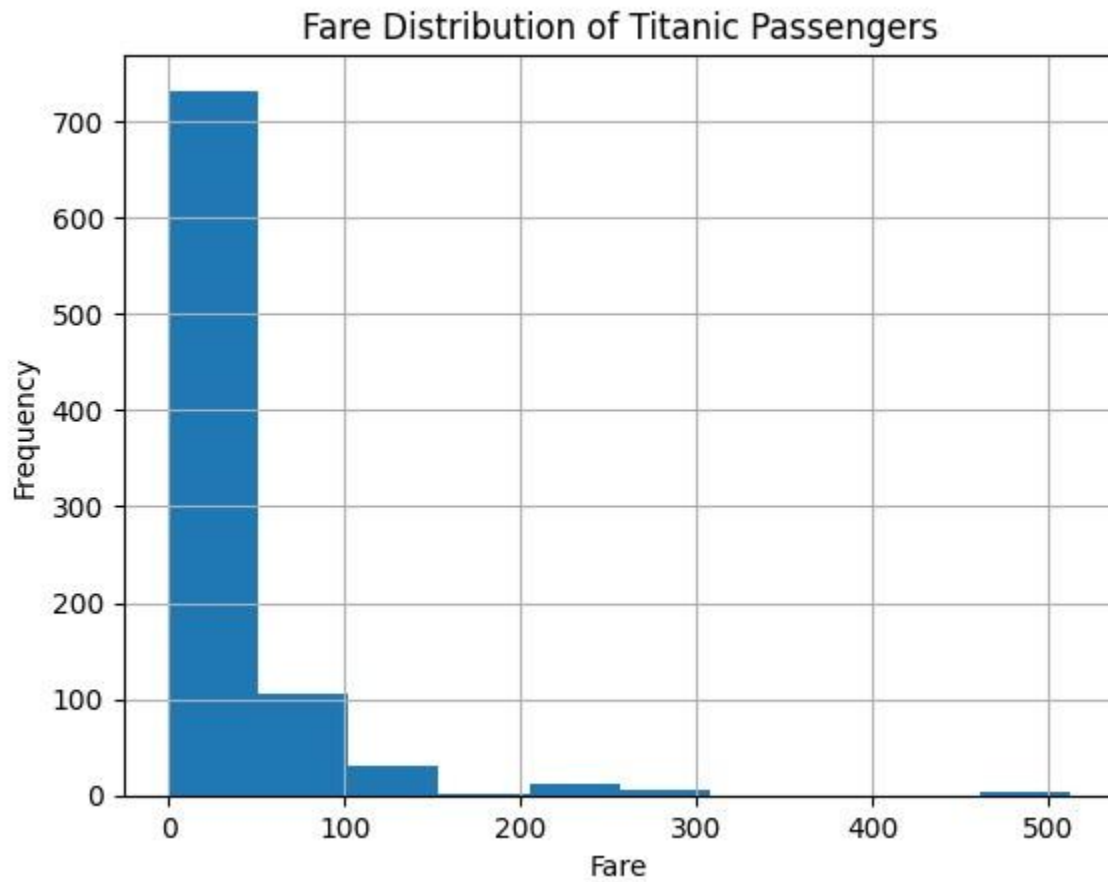
Perform data visualization, histogram and multiple variable summaries on titanic dataset

Program:

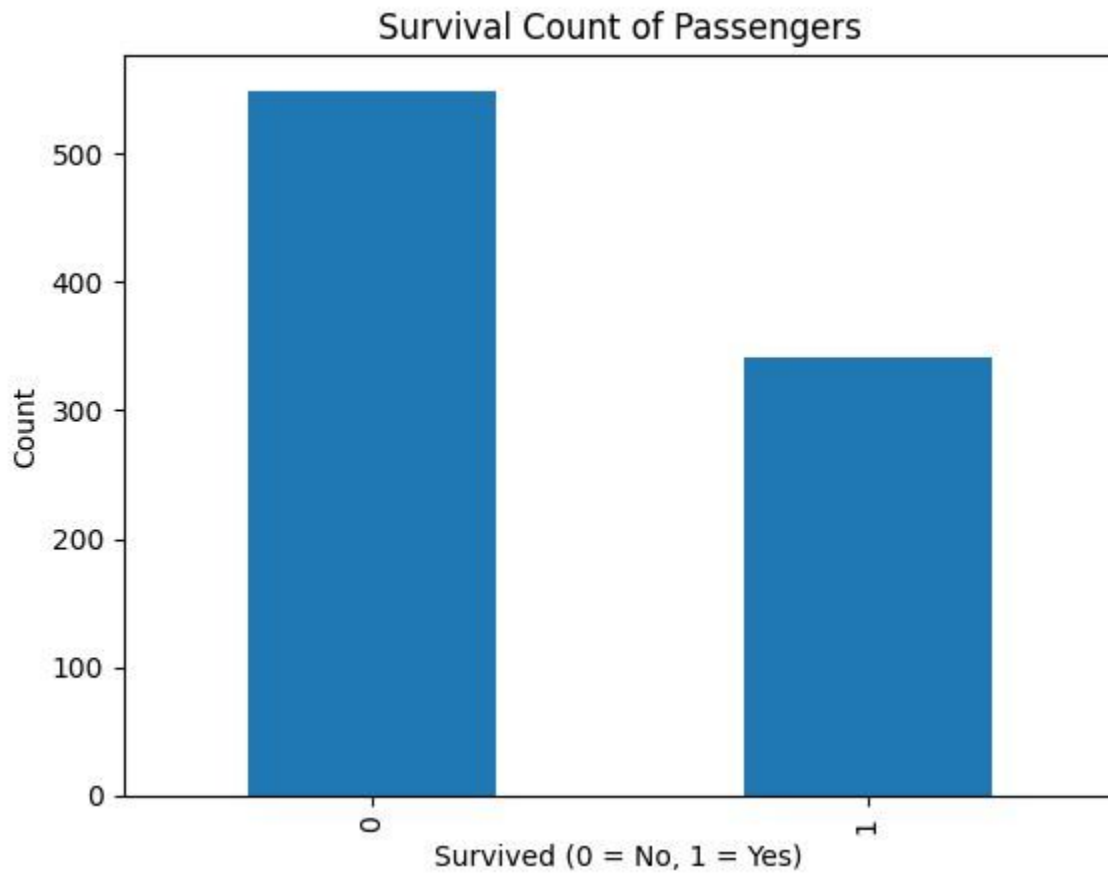
```
import pandas as pd import matplotlib.pyplot as plt
df = pd.read_csv("/content/titanic.csv")
print(df.info()) print(df.describe()) plt.figure()
df['Age'].dropna().hist() plt.xlabel("Age")
plt.ylabel("Frequency") plt.title("Age Distribution
of Titanic Passengers") plt.show()
```



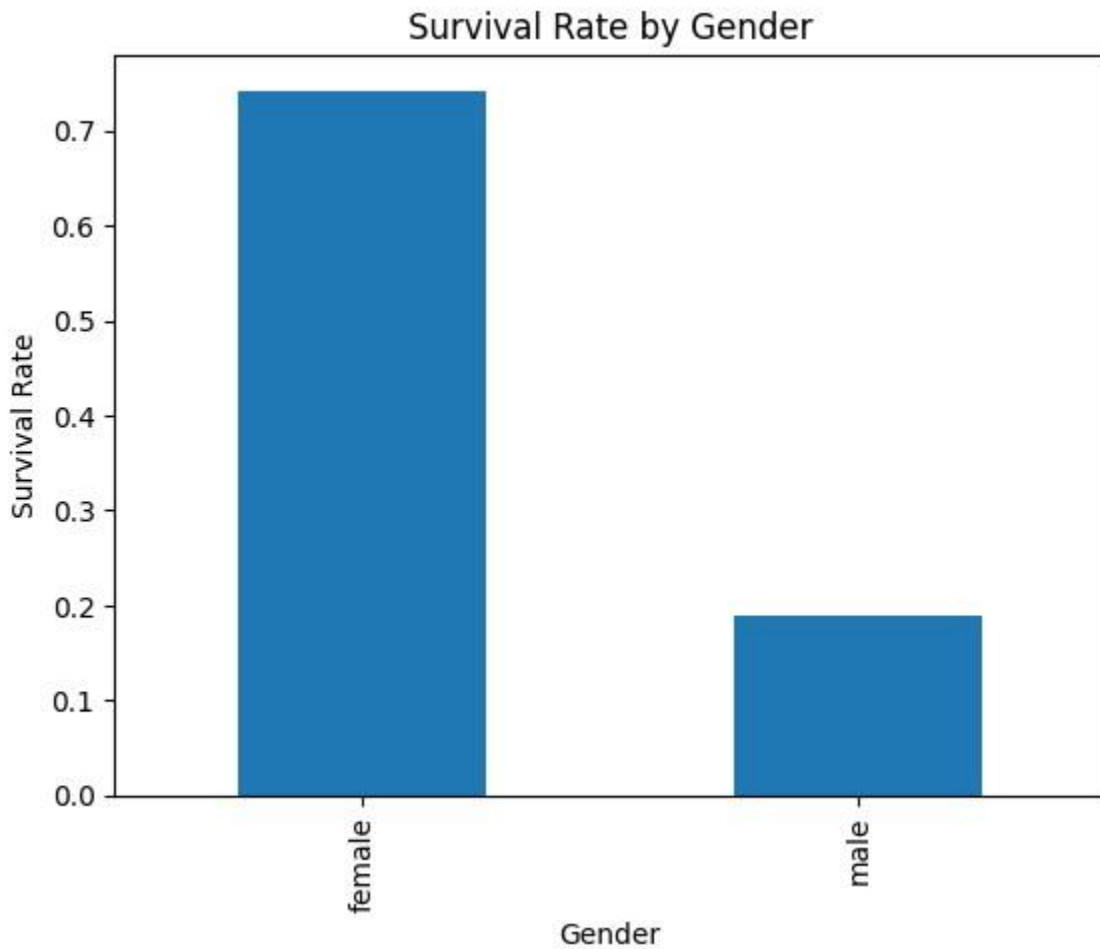
```
plt.figure() df['Fare'].hist() plt.xlabel("Fare")
plt.ylabel("Frequency") plt.title("Fare Distribution
of Titanic Passengers") plt.show()
```



```
plt.figure()  
df['Survived'].value_counts().plot(kind='bar')  
plt.xlabel("Survived (0 = No, 1 = Yes)")  
plt.ylabel("Count") plt.title("Survival Count  
of Passengers") plt.show()
```



```
gender_survival = df.groupby('Sex')['Survived'].mean()
plt.figure() gender_survival.plot(kind='bar')
plt.xlabel("Gender") plt.ylabel("Survival Rate")
plt.title("Survival Rate by Gender") plt.show()
```



```
class_survival = df.groupby('Pclass')['Survived'].mean()
plt.figure() class_survival.plot(kind='bar')
plt.xlabel("Passenger Class") plt.ylabel("Survival
Rate") plt.title("Survival Rate by Passenger Class")
plt.show()
```

